

Source: 0606 / s21 / qp_23 — Question 4

- 4 Variables x and y are such that $y = \frac{\sin x}{\cos x}$. Using differentiation, find the approximate change in y as x increases from $-\frac{\pi}{4}$ to $h - \frac{\pi}{4}$, where h is small. [4]

(b) Hence find the area enclosed between the curve $y = 2x^2 - 17x + 21$ and the x -axis. [3]